AirPulse

An Integrated Health Risk Assessment System

Combining smartphone-based PPG heart rate monitoring with real-time air quality data and personal health profiling for context-aware cardiovascular risk assessment.

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PPG SIGNAL PROCESSING | AIR QUALITY (AQI) | HEART RISK ASSESSMENT

Problem Statement

Current health monitoring systems analyze either physiological data (like heart rate) or environmental factors (like air quality) independently. However, real-world health risk depends on a combination of both internal body signals and external environmental conditions. Additionally, most systems do not consider personal health profiles such as age, exercise habits, or medical conditions. This leads to incomplete and non-personalized health assessment.

Therefore, there is a need for a system that integrates physiological sensing, environmental data, and user-specific information to provide a more accurate and context-aware health risk assessment.

Physiological data analyzed in isolation (heart rate only)	Environmental factors not linked to health risk (air quality ignored)	Personal health profiles not considered (age, habits, conditions)

A unified system integrating physiological sensing, environmental data, and personal health profiling is needed for accurate, context-aware risk assessment.

User Personas

Who Uses AirPulse?

Rohan, Gym-going Student

Age: 21 | City: Delhi

Situation: Runs outdoors, HR normal but Delhi AQI ~180. Exercises without knowing air quality risk.

Pain Point: Tracks fitness, ignores environment.

AirPulse Result: HR ✔️ Normal | AQI ❌ Poor → Risk: Moderate

Outcome: Shifts workouts indoors.

Ankit, Corporate Employee

Age: 30 | City: Gurgaon

Situation: Feels tired after work. HR slightly elevated, AQI moderate-high. Sedentary lifestyle.

Pain Point: Can't tell if it's stress, pollution, or something serious.

AirPulse Result: HR slightly high + AQI moderate + sedentary → Risk: Moderate

Outcome: Reduces outdoor exposure, adds light activity.

Neha, Working Mother

Age: 45 | City: Noida

Situation: Has mild BP issues. Plans morning walk. AQI: 210, HR slightly elevated.

Pain Point: Cautious but lacks clear decision support.

AirPulse Result: Risk: High → Avoid outdoor activity.

Outcome: Makes a safe decision immediately.

Mr. Sharma, Elderly Retired User

Age: 67 | City: Delhi

Situation: Son installed app. HR ~48 BPM (slightly low), AQI moderate. On heart medication.

Pain Point: Cannot interpret medical numbers easily.

AirPulse Result: Risk: Moderate → Simple message: "Take rest, avoid exertion."

Outcome: Actionable guidance without medical jargon.

Motivation

The motivation for this project comes from the increasing impact of air pollution on cardiovascular and respiratory health. Studies show that higher $PM_{2.5}$ levels significantly increase the risk of heart diseases. At the same time, smartphone-based PPG allows easy and accessible heart rate monitoring without external devices. However, these two aspects are rarely combined. This inspired us to build a system that bridges the gap between physiological monitoring and environmental awareness.



Air Pollution & Cardiovascular Risk

Rising $PM_{2.5}$ levels are strongly linked to increased risk of heart disease and respiratory conditions



Accessible PPG Monitoring

Smartphone-based PPG enables hardware-free heart rate monitoring — no external devices needed



Bridging the Gap

Combining physiological sensing with environmental awareness for a holistic, context-aware health system

Methodological Foundation

The methodology used in this work is inspired by the paper "Determination of Heart Rate from Photoplethysmogram using Fast Fourier Transform" by N. H. Mohd Sani et al., published in 2015. The paper proposes a frequency-domain approach for heart rate estimation, where the PPG signal is transformed using Fast Fourier Transform (FFT) and the dominant frequency component is used to compute heart rate in beats per minute.

> "Determination of Heart Rate from Photoplethysmogram using Fast Fourier Transform"

The following slides walk through each stage of this methodology — from raw PPG signal acquisition through to final heart rate estimation in BPM.



Signal Acquisition (PPG Capture)

Technique: Photoplethysmography (PPG)

The system begins with photoplethysmography (PPG) signal acquisition using the smartphone camera and flashlight. When the user places their finger over the camera lens, the flash illuminates the skin, and the camera captures subtle variations in light intensity caused by changes in blood volume with each heartbeat. These variations occur due to the pulsatile nature of blood flow in the fingertip.

Key Points:

- Captured video frames act as a source of physiological data
- Optical sensing: light absorption and reflection detect biological signals
- No external hardware required — highly accessible
- Output: raw signal embedded in video frames containing heart activity data (with noise to be filtered in next steps)

2 — Signal Extraction & Preprocessing

Technique: Red Channel Intensity Extraction + Band-Pass Filtering

The raw video data is converted into a usable signal through frame-wise intensity extraction. The red channel of each frame is analyzed because it is most sensitive to blood flow changes. The average intensity of the red channel is computed for every frame, forming a temporal waveform that represents the PPG signal over time.

Step 1 — Red Channel Extraction

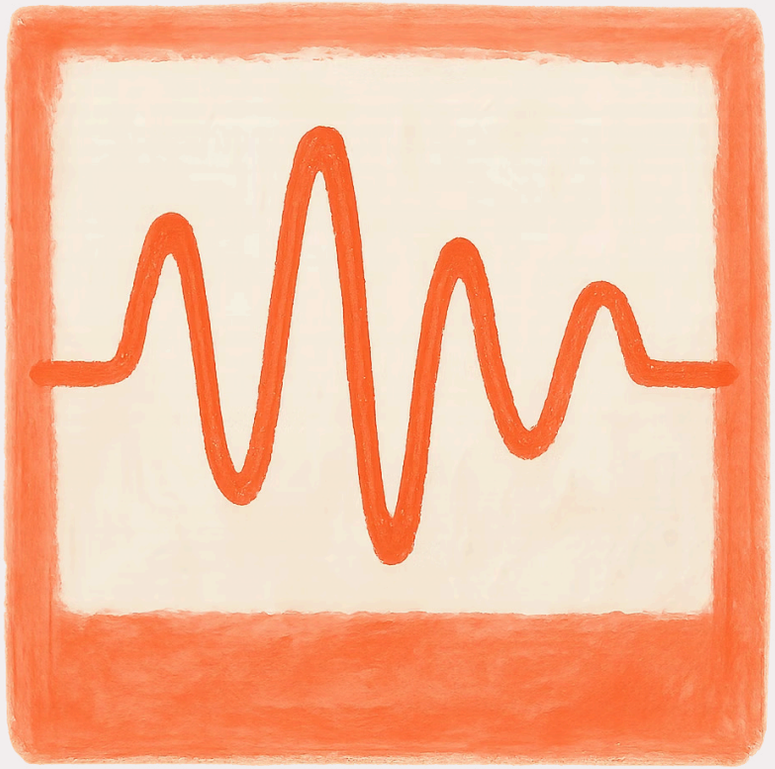
Average red channel intensity computed per frame to form a temporal waveform

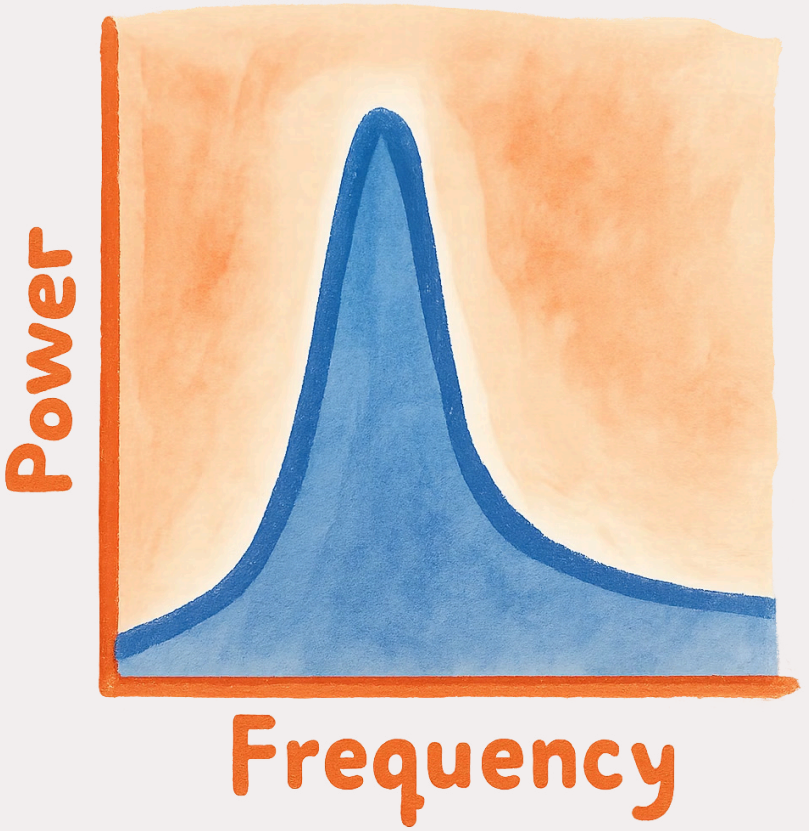
Step 2 — Noise Reduction & Smoothing

Removes disturbances from finger movement, lighting inconsistencies, and sensor limitations

Step 3 — Band-Pass Filtering (0.7–4 Hz)

Retains only the frequency range relevant to heart rate, producing a clean, stable signal ready for analysis





3 — Frequency Domain Transformation

Technique: Fast Fourier Transform (FFT)

After obtaining a clean PPG waveform, the system performs frequency-domain analysis using the Fast Fourier Transform (FFT). FFT converts the signal from the time domain into the frequency domain, revealing how frequently the signal repeats over time.

Why FFT over peak detection? More robust in noisy conditions — focuses on frequency patterns rather than time-based variations

Isolates the true heart rate signal even when the waveform is not perfectly clear

Core of the heart rate estimation process

4 — Heart Rate Estimation

Technique: Dominant Spectral Peak Detection + Frequency-to-BPM Conversion

Once the frequency spectrum is obtained, the system identifies the dominant frequency component corresponding to the user's heartbeat.

01

Step 1 — Dominant Spectral Peak Detection

Locate the frequency with the highest amplitude in the power spectrum — this corresponds to the heartbeat frequency

02

Step 2 — Frequency in Hertz (Hz)

The dominant peak is measured in Hz, representing cycles per second (heartbeats per second)

03

Step 3 — Frequency-to-BPM Conversion

Multiply the dominant frequency by 60 to convert from Hz to beats per minute (BPM)

04

Step 4 — Output & Integration

The calculated BPM is displayed to the user and fed into the Risk Assessment module as a key health input

This completes the physiological signal processing pipeline — delivering a reliable, hardware-free measure of cardiovascular activity.



Research Motivation

Air Pollution and Cardiovascular Health

> The research paper "Air Pollution and its Impact on Cardiovascular Health – It's Time to Act Fast!" was published in 2021 in the Indian Heart Journal by Rakesh Yadav, Surender Deora, and Geetika Yadav.

RAKESH YADAV · SURENDER DEORA · GEETIKA YADAV | INDIAN HEART JOURNAL, 2021

$PM_{2.5}$ & CVD

Air pollution, especially $PM_{2.5}$, is a major contributor to cardiovascular diseases

Increased Blood Pressure

Exposure leads to elevated blood pressure and heart stress

Arrhythmias & Heart Attack Risk

Linked to arrhythmias and higher risk of heart attack and stroke

Mortality & Hospitalization

Even small increases in $PM_{2.5}$ significantly raise cardiovascular mortality risk and hospital admissions



Final Risk Assessment Logic

Overall Risk Score Formula

$$RiskScore = S_{HR} + S_{AQI} + S_{Resp} + S_{Cardio} + S_{Age} + S_{Exercise}$$

Total Score ∈ [0, 10]

1. Heart Rate Score (S_HR)

Condition	Score
50 ≤ HR ≤ 100	0
45 ≤ HR < 50 or 100 < HR ≤ 110	2
HR < 45 or HR > 110	3

2. AQI Score (S_AQI)

Condition	Score
AQI ≤ 50	0
51 ≤ AQI ≤ 100	1
101 ≤ AQI ≤ 200	2
AQI > 200	3

3. Respiratory Condition (S_Resp)

Condition	Score
No	0
Yes	1

4. Cardiovascular Condition (S_Cardio)

Condition	Score
No	0
Yes	2
Yes (on medication)	3

5. Age Factor (S_Age)

Condition	Score
Under 25	0
25–40	1
40–60	2
> 60	3

6. Exercise Frequency (S_Exercise)

Condition	Score
> 3 times/week	0
1–3 times/week	1
Rarely/Never	2

Final Risk Classification

Low Risk

Score 0–3

Moderate Risk

Score 4–6

High Risk

Score 7–10

Example: HR = 47, AQI = 73

Score = 2 (HR) + 1 (AQI) + 0 + 0 + 1 = 4

→ Moderate Risk

Getting Started: Profile & Dashboard

Step 1 — User Profile Setup

On first launch, AirPulse asks you to enter a few key health details to personalise your experience. These include:

- Exercise frequency
- Respiratory conditions
- Cardiovascular conditions
- Age range

This one-time setup ensures every risk assessment is tailored specifically to your health profile.

Step 2 — Dashboard at a Glance

After setup, the dashboard gives you an instant overview of your current health state, displaying:

- Current heart rate
- Current AQI (Air Quality Index)
- Current risk level

From here, three quick-access shortcuts take you directly to the app's core features: **Scan Heart Rate**, **Air Quality View**, and **Risk Assessment**.



Heart Rate & Air Quality Monitoring

📷 Heart Rate Measurement (PPG)

Selecting **Scan Heart Rate** activates the smartphone camera and flashlight. You simply place your fingertip over the lens, and the app captures the video signal from your fingertip. Using **Photoplethysmography (PPG)** analysis, it estimates your heart rate in BPM. On-screen guidance helps you keep your finger steady for a reliable reading.

🌍 Real-Time AQI Monitoring

AirPulse uses your device's location to fetch live Air Quality Index data. The app displays the **AQI value and category**, your location name, and detailed pollutant breakdowns including **PM2.5, PM10, NO_2, O_3, and CO**. This gives you a clear, immediate picture of the air you're breathing and the environmental conditions around you.

Risk Assessment & Recommendations



Risk Assessment

AirPulse combines your heart rate, AQI data, and health profile to calculate a personalised risk level — **Low**, **Moderate**, or **High**. A numeric risk score is shown alongside the key contributing factors so you always understand what's driving the result.



Smart Recommendations

Based on your risk level, AirPulse delivers short, practical, actionable advice — such as reducing outdoor activity, consulting a physician if symptoms arise, taking precautions during poor air quality, or maintaining light exercise when conditions are safe.

History Tracking & Settings

📊 History Tracking

Every assessment is automatically saved in the **History** section, giving you a personal log of past readings. You can review:

- Heart rate readings over time
- AQI values at each session
- Risk results with date and time stamps

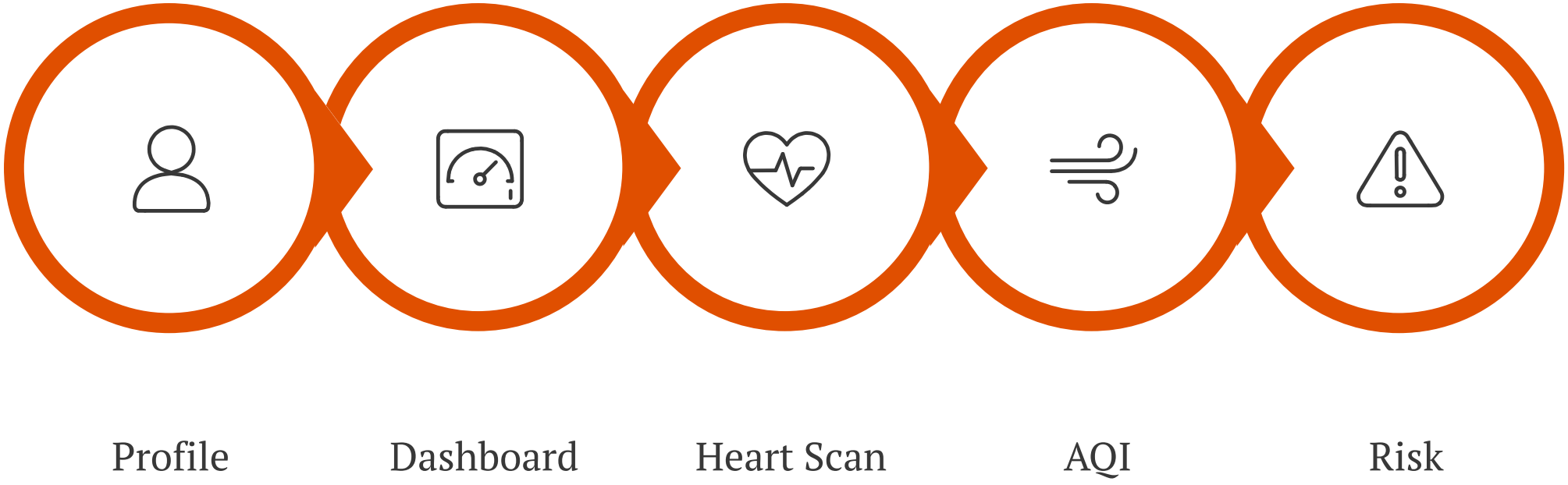
This transforms AirPulse from a live monitor into a powerful **long-term health tracking tool**, helping you spot patterns and trends in your wellbeing.

⚙️ Settings & Preferences

The Settings screen keeps the app flexible and personalised. Users can:

- Manage and update their health profile
- Enable or disable location access for AQI
- View account details
- Access app version information

These controls ensure AirPulse adapts to each individual user's needs and privacy preferences.



The complete AirPulse workflow is designed to be intuitive at every step — from first-time setup through to ongoing daily monitoring and long-term health trend analysis.

Activity Recognition Chain (ARC)

How AirPulse Processes Raw Data into Health Decisions

ARC is a step-by-step pipeline that converts raw sensor data into meaningful decisions. In AirPulse: Raw Data → Processing → Understanding → Action

Sensing

- Heart signal via camera PPG
- AQI via location-based API
- User profile via manual input

Pre-processing

- Remove noise from PPG signal
- Smooth intensity values
- Filter irrelevant frequencies

Feature Extraction

- Apply FFT to PPG signal
- Extract dominant frequency
- Convert to heart rate (BPM)

Inference / Decision

- Combine HR + AQI + health profile
- Apply risk scoring model
- Output: Low / Moderate / High risk

Action / Output

- Show risk level to user
- Provide personalized recommendations
- Store reading in history

ARC maps PPG, AQI, and user data into a personalized health risk output — converting raw sensor signals into actionable health decisions.

Design Thinking Process

A User-Centered Approach to AirPulse

Empathize

Users don't understand raw HR or AQI data

Decisions based on assumptions ("I feel fine")

Pollution impact often ignored

Need for simple, actionable insights

Focus: reduce cognitive load

Define

Existing systems analyze health in isolation

No integration of HR + AQI + personal profile

Leads to incomplete risk understanding

Need: context-aware health monitoring system

Ideate

Combine PPG heart rate + AQI + user profile

Rule-based model chosen for interpretability

Single risk score + category output

Accessibility focus: no extra hardware needed

Prototype

Built AirPulse mobile app

Modules: PPG heart rate (camera), AQI (real-time), user profile, risk engine, dashboard

Test

Tested multiple HR + AQI combinations

Verified logical risk outputs

Iteratively refined scoring + UI

Ensured easy interpretation by all users

Followed a user-centered design process: Empathize → Define → Ideate → Prototype → Test

Design Principles

Building for Simplicity, Clarity & Context-Awareness

Simplicity

Used Low/Moderate/High instead of raw numbers. Users understand results instantly.

Clarity & Interpretability

Clear labels (HR, AQI, Risk). Direct meaning: "What should I do?"

Context-Awareness

Combines physiological + environmental + personal data for meaningful output.

Personalization

Risk adapts to age, health conditions, and lifestyle. Not generic.

Accessibility

Uses smartphone camera for PPG. No external hardware needed.

Real-Time Feedback

Instant HR + AQI + risk with immediate recommendations.

Actionable Insights

Goes beyond showing data — gives users clear next steps.

Consistency

Same structure across all screens. Reduces confusion.

Low User Effort

Minimal input, one-tap measurement. Smooth experience.

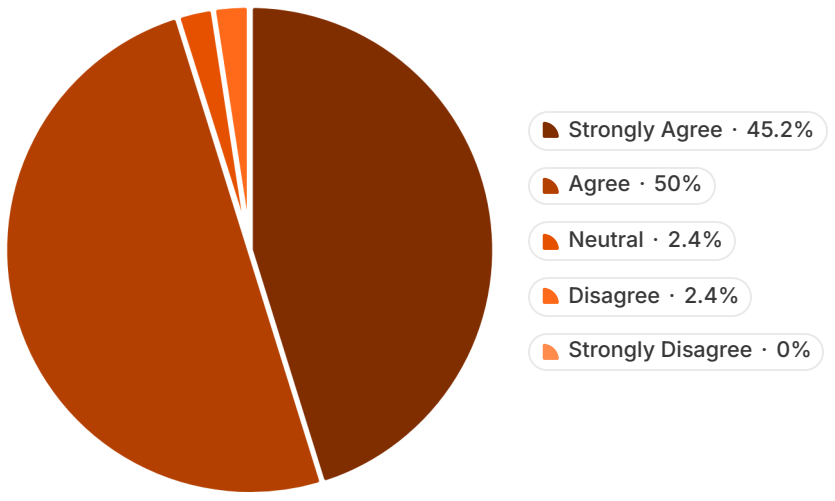
We focused on reducing cognitive load, increasing interpretability, and making the system context-aware and personalized — rather than just data-driven.

UI Feedback Survey Results

42 Responses — User Experience Evaluation

A UI feedback survey was conducted with 42 users to evaluate the usability, clarity, and overall experience of AirPulse.

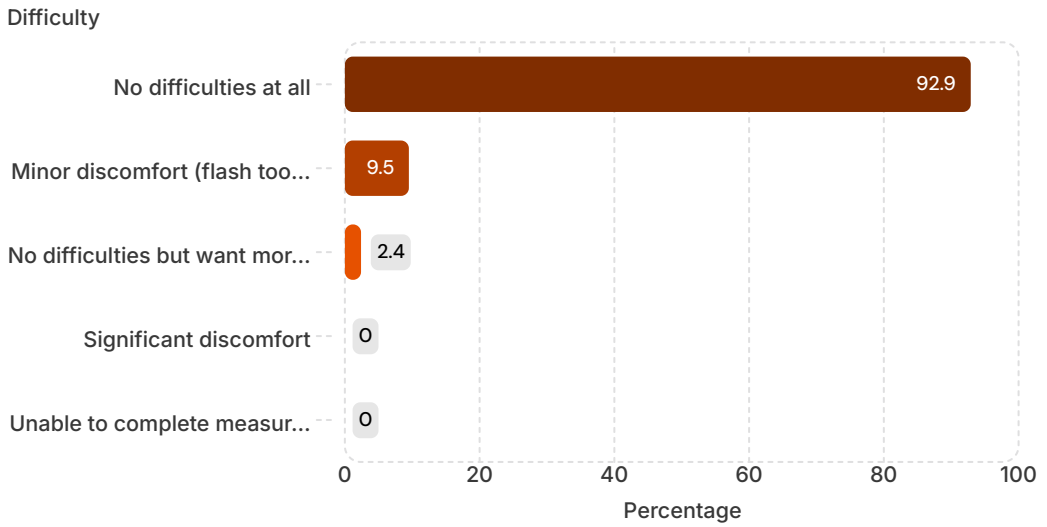
Overall Design Simplicity



The overall design of the app feels simple and not overwhelming.

Key stat: 95.2% agreed or strongly agreed

Physical Comfort During Measurement



Did you face any physical difficulties while measuring heart rate using the camera and flash?

User Feedback Highlights

> "Easy and very accessible"

> "The app is really useful and simple to use. An improvement to heart rate features would be great — for example, taking inputs of the user's latest activity and comparing normal heart rate to current heart rate to better determine risk."

92.9% of users experienced no difficulties — validating AirPulse's hardware-free, accessible design approach.

Future Improvements & Limitations

Where AirPulse Goes Next

Future Enhancements 🚀

Activity-Aware Monitoring (from user feedback)

Users want the system to differentiate between Walking, Running, and Resting.

Planned: Use inbuilt smartphone accelerometer to detect activity in real time and adjust risk assessment accordingly.

→ Makes the system more context-aware and accurate.

Current Limitations ⚠️

- PPG Accuracy — Sensitive to finger movement and lighting conditions
- Rule-Based Model — Does not use a trained ML/DL model
- Limited Sensing — Physical activity not yet considered
- AQI Limitation — Based on location data; may not reflect micro-environment
- Not a Medical Tool — Provides indicative risk, not clinical diagnosis

While our prototype is fully functional, future improvements focus on activity recognition and enhancing accuracy through better sensing and modeling.

Real-World Testing of AirPulse

Study Setup 🔬

- Participants: 10 users (students + working professionals)
- Duration: 2–3 minutes per reading
- Reference device: Oximeter
- Conditions tested: Resting, Light activity (after walking), Different AQI levels

AQI Integration 🌫️

AQI values matched API data accurately

Limitation: May differ from actual immediate surroundings

Heart Rate Accuracy (PPG) 📈

±4–7 BPM

Average error at rest

±8–12 BPM

Error under movement/noise

Accuracy improves when finger is stable and lighting is consistent.

Risk Assessment Validation ⚖️

Logical Variation: Risk levels varied logically across scenarios

Consistent Behavior: Higher HR + Higher AQI → Higher Risk

User Feedback

80%

Found output easy to understand

70%

Preferred risk categories over raw numbers

⚠ The system performs reliably for real-world use, but accuracy depends on signal quality and environmental conditions.

Core Concepts Larned

Context-Aware Computing

Combines heart rate + AQI + user profile to deliver personalized health risk — not a generic output.

Key idea: The system adapts its output based on who the user is and what their environment looks like.

Multimodal Data Fusion

Integrates three distinct data streams:

- Physiological data (PPG heart rate)
- Environmental data (AQI)
- Personal data (age, conditions, lifestyle)

Key idea: Converts multiple inputs into one meaningful, interpretable health insight.

AirPulse is not just a sensor app — it is a context-aware, multimodal health intelligence system.

Thank You :)

Questions & Discussion

We're happy to answer any questions about AirPulse!

AREEB NIKHIL RISHAB

AirPulse — Bridging Physiological Sensing & Environmental Awareness for Smarter Health Decisions